Flow Matching Cheat Sheet

PART 1: The SDE Framework (Continuous Diffusion)

1. The Forward SDEs (VE and VP)

The Formulas:

$$\textbf{VE-SDE (NCSN):}\ dx_t = \sqrt{\frac{d\sigma^2(t)}{dt}} dW_t$$

$$\textbf{VP-SDE (DDPM):}\ dx_t = -\frac{1}{2}\beta(t)x_t dt + \sqrt{\beta(t)} dW_t$$

Variables:

- $f(x,t)$: The drift coefficient (the dt multiplier). For VE, $f = 0$. For VP, $f = -\frac{1}{2}\beta(t)x_t$.
- $g(t)$: The diffusion coefficient (the dW_t multiplier).

When to use it:

- "Identify the drift and diffusion coefficients for the given model."
- "Write the continuous-time limit of the DDPM forward process."
- "What is the deterministic pull $f(x,t)$ applied to the particle at this step?"

2. The Reverse SDE

The Formula:

$$dx_t = [f(x,t) - g^2(t)\nabla_x \log P_t(x)]dt + g(t)d\bar{w}_t$$

Variables:

- $\nabla_x \log P_t(x)$: The true score (which you replace with S_θ for calculations).
- $d\bar{w}_t$: Reverse-time Brownian motion (random noise).

When to use it:

- "Write the continuous equation governing the backward generative process."
- "Given the drift and diffusion, what is the exact Reverse SDE?"
- *Trap Warning:* This is purely continuous. If you are given a finite time step Δt and asked to calculate an actual number, you must use the discretized Euler-Maruyama formula below.

PART 2: Discretized Samplers (The Predictor-Corrector)

3. The Predictor Step (Euler-Maruyama Discretization)

The Formula:

$$x_{i-1} = x_i - [f(x_i, t_i) - g^2(t_i)S_\theta(x_i, t_i)]\Delta t + g(t_i)\sqrt{\Delta t}Z_i$$

Variables:

- Δt: The finite time step size.
- Z_i: A sampled standard normal noise vector $\mathcal{N}(0, I)$.

When to use it:

- "Calculate the state of the image at the next step x_{i-1} using Euler-Maruyama."
- "Execute one Predictor step for the Reverse SDE."
- "Given the network score S_θ and time step Δt, compute the numerical update."

4. The Probability Flow ODE (Continuous)

The Formula:

$$\frac{d}{dt}x_t = f(x, t) - \frac{1}{2}g^2(t)\nabla_x \log P_t(x)$$

Variables:

- Notice the critical $\frac{1}{2}$ multiplier attached to the diffusion/score term.

When to use it:

- "What is the deterministic ODE that shares the same marginals as the SDE?"
- "Write the continuous Probability Flow equation."

5. The ODE Sampler Step (Discrete)

The Formula:

$$x_{i-1} = x_i - \left(f(x_i, t_i) - \frac{1}{2}g^2(t_i)S_\theta(x_i, t_i)\right)\Delta t$$

Variables:

- There is **NO** random Z_i noise term added at the end.

When to use it:

- "Calculate the next step using the deterministic ODE solver."
- "Perform a step of probability flow generation without injecting randomness."
- *Trap Warning:* If the question says "ODE" or "Deterministic," you must halve the score term and drop the Z noise term entirely.

PART 3: Standard Flow Matching

6. Linear Interpolation (Particle Position)

The Formula:

$$X_t = tX_1 + (1 - t)X_0$$

Variables:

- t: Time parameter where $t = 0$ is pure noise and $t = 1$ is clean data.
- X_0: The initial pure noise vector.
- X_1: The final clean data vector.

When to use it:

- "Calculate the exact position of the particle at time t."
- "Determine the intermediate state between this noise point and data point."
- "What is the input to the velocity network v_θ at this time step?"

7. Standard Flow Target Velocity

The Formula:

$$v = X_1 - X_0$$

Variables:

- A constant vector representing the straight-line trajectory.

When to use it:

- "Calculate the target velocity for this linear interpolation path."
- "What is the optimal vector the network should predict for standard flow matching?"
- "Compute v_{known} for this noise-data pair."

PART 4: Conditional Flow Matching (CFM)

8. The General Conditional Interpolant

The Formula:

$$X_t = m(t, z) + a(t)\epsilon$$

Variables:

- $m(t, z)$: The deterministic mean path equation.
- $a(t)$: The noise scale equation.
- ϵ: Standard Gaussian noise.

When to use it:

- "Given a custom mean path and noise scale, calculate the position of the noisy particle."
- "Determine the conditional distribution $P_t(x|z)$."

9. The Master Conditional Target Velocity

The Formula:

$$V_t(X_t|z) = \dot{m}(t, z) + \frac{\dot{a}(t)}{a(t)}(X_t - m(t, z))$$

Variables:

- $\dot{m}(t, z)$: The time-derivative of the mean function.
- $\dot{a}(t)$: The time-derivative of the noise scale function.
- X_t: The current particle position.

When to use it:

- "Calculate the exact conditional target velocity $V_t(X_t|z)$."
- "What vector should the CFM network be penalized against given this custom interpolant?"
- "Given X_t, compute the velocity field for this specific latent pair."
- *Trap Warning:* You must take the derivatives of m and a with respect to t before plugging in any numerical time values.

10. Optimal Transport (OT) Cost Matrix

The Formula:

$$C_{ij} = \|X_0^i - X_1^j\|_2^2$$

Variables:

- X_0^i: The i-th noise vector in your mini-batch.
- X_1^j: The j-th data vector in your mini-batch.

When to use it:

- "Construct the cost matrix for OT coupling."
- "Calculate the pairwise distances between the noise batch and data batch."
- "Determine the optimal pairings for this mini-batch."

PART 5: Mean-Flow Models

11. The Mean Flow Identity (Target Velocity)

The Formula:

$$V_{target} = v(x_t, t) + (t_e - t)\frac{d}{dt}v_\theta(x_t, t, t_e)$$

(Which is practically computed via JVP as:)

$$V_{target} = (X_1 - X_0) + (t_e - t) \cdot JVP$$

Variables:

- $v(x_t, t)$: The instantaneous target velocity (usually $X_1 - X_0$).
- t_e: The targeted end time (often 1.0).
- t: The current start time.
- JVP: The Jacobian-Vector Product supplied by the automatic differentiation engine.

When to use it:

- "Calculate the target *average* velocity V_{se} for the Mean-Flow network."
- "Using the provided JVP, find the ground-truth vector for one-step generation."
- "Apply the Mean Flow Identity to compute the training target."

12. Mean-Flow Training Loss

The Formula:

$$L_{MF}(\theta) = E\left[\|v_\theta(x_t, t, t_e) - \text{StopGradient}(V_{target})\|_2^2\right]$$

Variables:

- $v_\theta(x_t, t, t_e)$: The average velocity predicted by your neural network.
- $\text{StopGradient}(V_{target})$: The target computed in Formula 11, strictly isolated from the optimizer.

When to use it:

- "Write the loss function used to train the Mean-Flow network."
- "Calculate the squared error for this one-step generation training step."
- *Trap Warning:* If the question asks you to write the formula, you **must** include the `StopGradient` explicitly, or it is mathematically invalid.